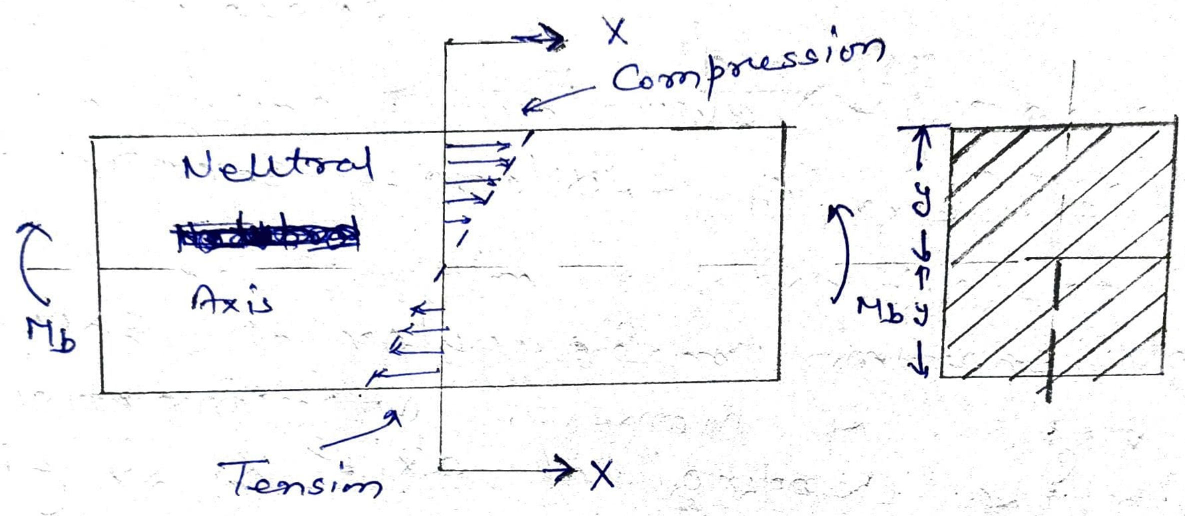
Unit-4

Design of Members in Bending:

Beams :-



(a) (b)

σ_b = Bending stress at a distance of y from the neutral axis (N/mm^2) or MPa.

M_b - Applied bending moment (N-mm).

I - Moment of Inertia of the cross-section about the neutral axis (mm^4)

Bending stress at any fibre

$$\sigma_b = \frac{M_b y}{I} \quad \text{— (1)}$$

for Rectangular cross-section

$$I = \frac{bd^3}{12} \quad \text{— (2)}$$

b = distance parallel to the neutral axis (mm).

d = distance perpendicular to the neutral axis (mm)

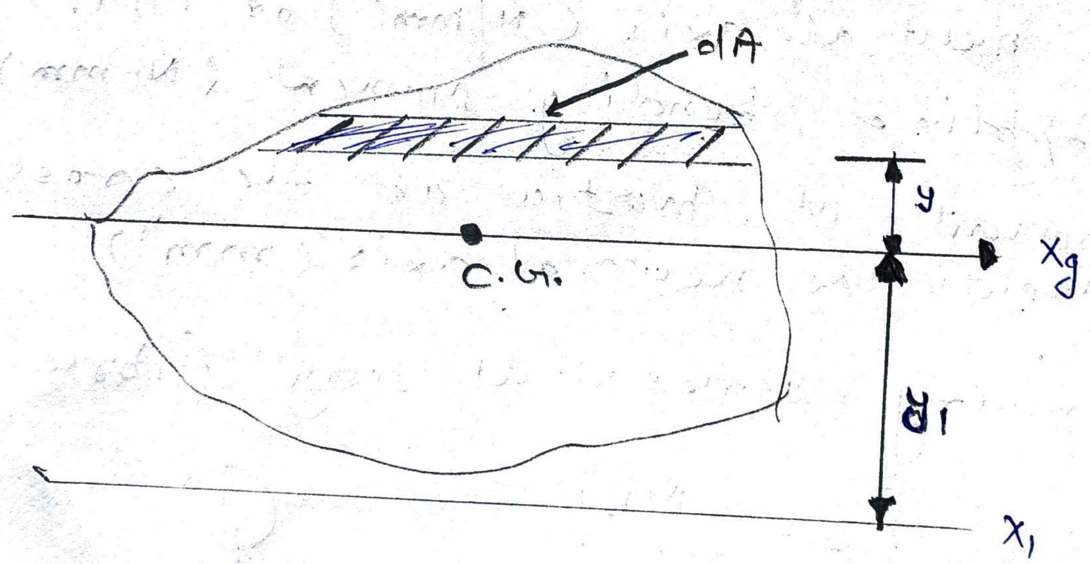
for a circular cross-section

$$I = \frac{\pi d^4}{64} \quad \text{—— (3)}$$

d: diameter of the cross-section

If x-section is irregular as shown in fig. the moment of inertia about the centroidal axis X_g is given by

$$I_{xg} = \int y^2 dA \quad \text{—— (4)}$$



The parallel-axis theorem for this area is given by the expression

$$I_{x1} = I_{xg} + Ay_1^2 \quad \text{—— (5)}$$

$I_{x_1} = \cancel{I_{xg} + Ay_1^2}$, Moment of inertia of the area about X_1 axis, which is parallel to the axis X_g and located at a distance y_1 from X_g.

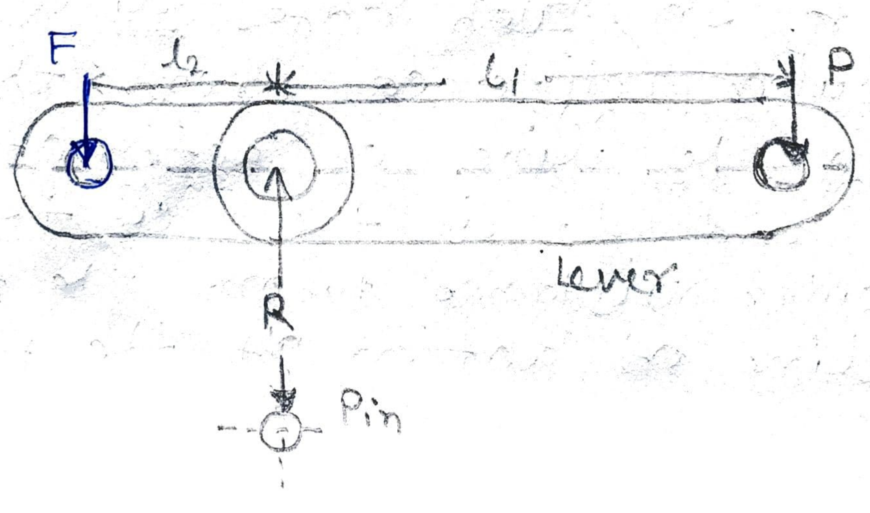
I_{xg} = Moment of inertia of the area about its own centroidal axis

A = Area of the x-section.

Levers:- A lever is defined as a mechanical device in the form of a rigid bar pivoted about the fulcrum to multiply or transfer the force.

The Length of the lever is decided on the basis of leverage required to exert a given load F by means of an effort P. The cross-section of the lever is designed on the basis of bending stresses. The design of the lever consists of the following steps.

Step:1 Force Analysis.



Effort required to produce force about the fulcrum.

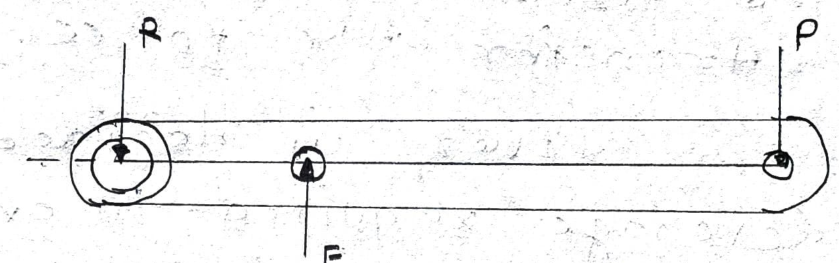
$$F \times l_2 = P \times l_1 \quad —(1)$$

$$P = F\,(l_2/l_1) \quad —(2)$$

according to freebody ~~fig~~ diagram 1 of force acting on the lever

$$\boxed{R = F + P} \quad —(3)$$

Reaction at the fulcrum Pin.



according to freebody diagram no. 2 force acting on the lever

$$F = R + P$$

$$\boxed{R = F - P} \quad —(4)$$

Both case are for parallel force. If the forces F and P acting inclined to one other. In such cases l_1 is the perpendicular distance from the fulcrum to the line of action of the force P. and l_2 is the is the perpendicular distance from the fulcrum to the line of action of the force F.

In that case

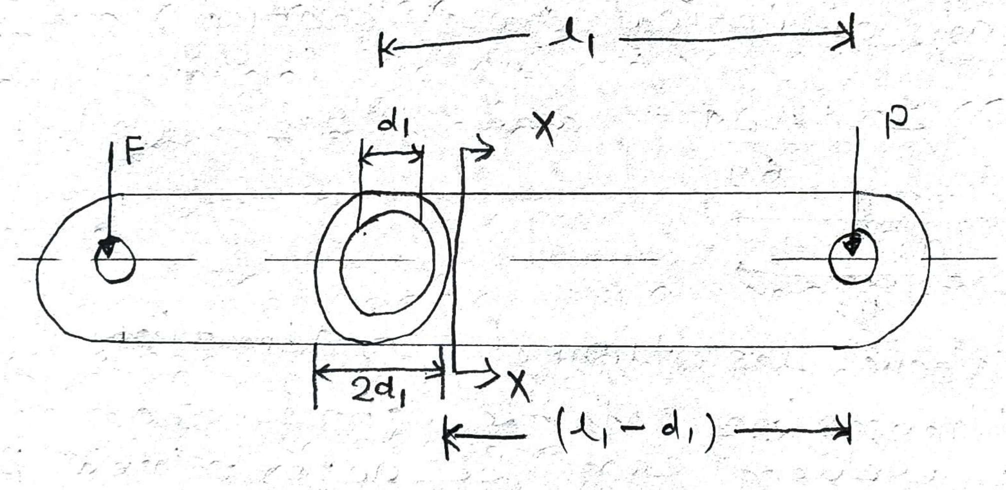
$$R = \sqrt{F^2 + P^2 - 2FP\cos\theta}$$

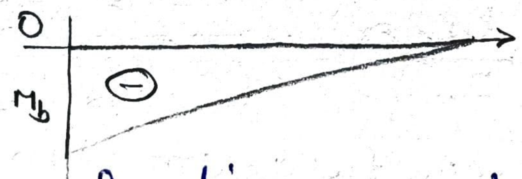
at right angle

$\theta = 90^\circ$ & $\cos\theta = 0$

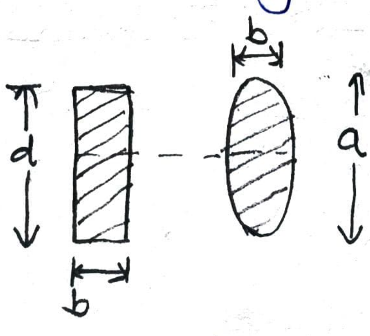
so $\boxed{R = \sqrt{F^2 + P^2}}$ —— (5)

Step-2 :- Design of Lever Arm !





Bending moment dig.



Section at XY.

Bending moment at section XX and it is given by,

$$M_b = P(l_1 - d_1) \quad — (6)$$

Cross-section of the lever can be rectangular, elliptical or I-section.

for rectangular cross-section

$$I = \frac{bd^3}{12} \quad \text{and} \quad y = d/2 \quad — (7)$$

where $\boxed{d = 2b}$

for an elliptical cross-section

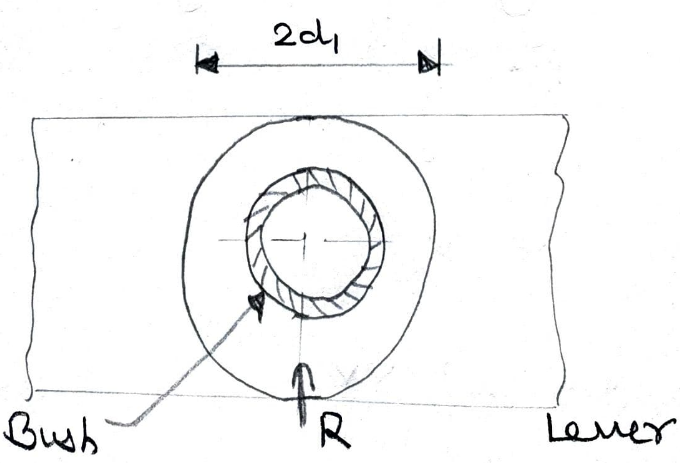
$$I = \frac{\pi b a^3}{64} \quad \text{and} \quad y = a/2 \quad — (8)$$

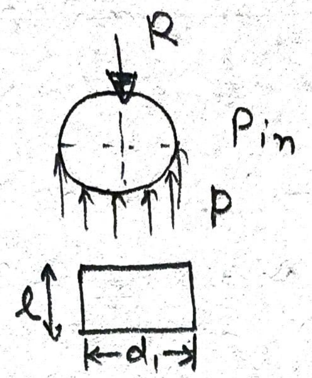
Where $\boxed{a = 2b}$

using above mentioned proportions the dimensions of the cross-section of the lever can be determined by

$$\sigma_b = \frac{M_b y}{I}$$

Step-3 Design of fulcrum Pin





Projected area of pin is
$(d_1 \times l_1)$

So,

$$R = p(d_1 \times l_1) \quad \text{— (9)}$$

$$p = \frac{R}{(d_1 \times l_1)} \quad \text{— (10)}$$

Similar Example of knuckle joint pin
where

$$\sigma_c = \frac{\text{force}}{\text{Projected area}}$$

$$\sigma_c = \frac{P}{(d \times l_1)} \quad \text{— (11)}$$

Laminated Springs:- A leaf spring is a form of spring commonly used for the suspension in wheeled vehicles. Originally it is called a Laminated spring.

For purpose of analysis It divide into two groups
(1) Master leaf along with graduated

Leaves
(2) Extra full length ~~leaf~~ leaves forming the other.

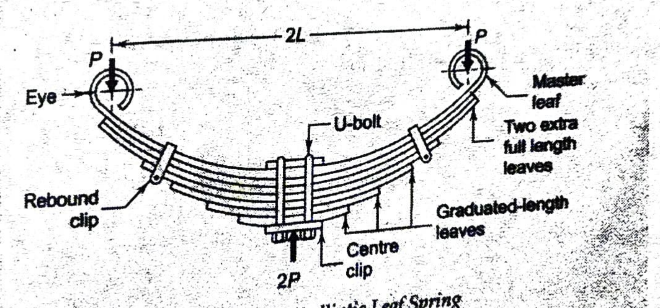


Fig. 10.31 *Semi-elliptic Leaf Spring*

n_f = no. of extra full length leaves

n_g = no. of graduated-length leaves ~~inclu~~ including master leaf

n = total no. of leaves

b = width of each leaf (mm)

t = thickness of each leaf (mm)

L = length of the cantilever or half the

P = force applied at the end of the spring (N)

P_f = Portion of P taken by the extra full-length leaves (N)

P_g = Portion of P taken by the graduated-length leaves (N)

Bending stress in the plate at the support.

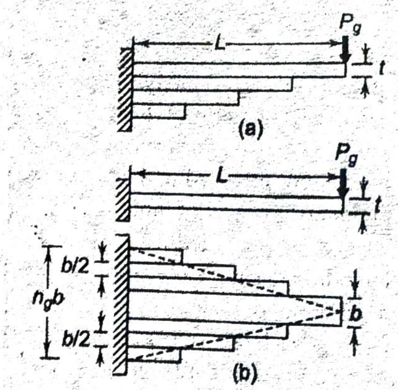


Fig. 10.32 *Graduated-length Leaves as Triangular Plate*

$$(\sigma_b)_g = \frac{M_b y}{I} = \frac{(P_g L)(t/2)}{\left[\frac{1}{12}(n_g b)(t^3)\right]}$$

$$(\sigma_b)_g = \frac{6 P_g L}{n_g b t^2} \quad \text{— (1)}$$

deflection (δ_g) at the load point of the triangular plate is given by

$$\delta_g = \frac{P_g L^3}{2 E I_{max}}$$

$$\delta_g = \frac{P_g L^3}{2E\left[\frac{1}{12}(n_g b)(t^3)\right]}$$

$$\delta_g = \frac{6 P_g L^3}{E n_g b t^3} \quad \text{— (2)}$$

Similarly, the extra full-length leaves can be treated as a rectangular plate of thickness t and uniform width ($n_f b$). So

$$(\sigma_b)_f = \frac{M_b y}{I} = \frac{(P_f L)(t/2)}{\left[\frac{1}{12}(n_f b)(t^3)\right]}$$

$$(\sigma_b)_f = \frac{6 P_f L}{n_f b t^2} \quad \text{— (3)}$$

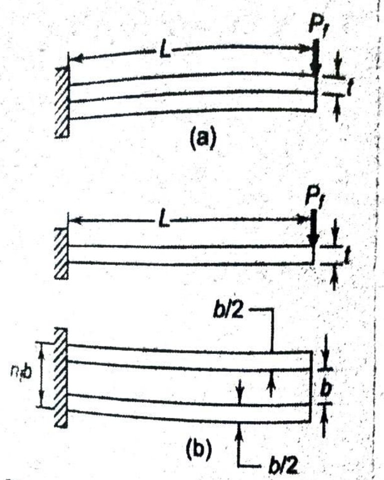


Fig. 10.33 Extra full-length leaves as rectangular plate

The deflection at the load point is given by

$$\delta_f = \frac{P_f L^3}{3EI} = \frac{P_f L^3}{3E\left[\frac{1}{12}(n_f b)(t^3)\right]} \quad — (4)$$

$$\delta_f = \frac{4 P_f L^3}{E n_f b t^3} \quad — (5)$$

Since the deflection of full-length leaves is equal to the deflection of graduated-length leaves.

$$\delta_g = \delta_f$$

$$\frac{6 P_g L^3}{E n_g b t^3} = \frac{4 P_f L^3}{E n_f b t^3}$$

$$\frac{P_g}{P_f} = \frac{2 n_g}{3 n_f} \quad — (6)$$

also $P_g + P_f = P$ — (7)

from eq^n (6) & (7)

$$P_f = \frac{3n_f P}{(3n_f + 2n_g)} \quad -8$$

$$P_g = \frac{2n_g P}{(3n_f + 2n_g)} \quad -9$$

Substituting the above values in eq^n 4 & 3

$$(\sigma_b)_g = \frac{12PL}{(3n_f + 2n_g)bt^2}$$

$$(\sigma_b)_f = \frac{18PL}{(3n_f + 2n_g)bt^2}$$

Deflection at the end of the spring

$$\delta = \frac{12PL^3}{Ebt^3(3n_f + 2n_g)} \quad -10$$